TECHNICAL PRODUCT REQUIREMENTS DOCUMENT

Online RFQ & Tendering Platform

Version 3.0 | February 2026

Developer-Ready Specification

Status: DRAFT | Classification: Confidential

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1. Overview & Purpose

This Technical Product Requirements Document (PRD) defines the complete specification for an Online RFQ & Tendering Platform. It is written to be developer-ready: every data model, state machine, permission rule, API contract hint, and non-functional constraint is documented so that engineering teams can plan sprints directly from this document.

The platform enables Buyers to float Requests for Quotation (RFQ) or formal Tenders — open to the market or limited to an invited set of vendors. Vendors submit structured technical and commercial bids. Evaluators assess compliance and pricing through a two-stage workflow. The entire lifecycle is auditable, tamper-evident, and extensible.

1.1 Document Conventions

- PK = Primary Key, FK = Foreign Key, UUID = v4 unique identifier.
- All monetary fields use NUMERIC(18,2) unless stated otherwise.
- All timestamps are TIMESTAMPTZ (UTC). Timezone conversion is a UI concern only.
- "Nullable" means the column permits NULL; all other columns are NOT NULL by default.
- Enum values are listed explicitly; the database enforces them via CHECK or a lookup table.

2. Goals & Non-Goals

2.1 Goals

- Digitize the full RFQ & Tender lifecycle: creation → publication → bidding → evaluation → award.
- Support structured specifications with selectable feature groups and hierarchical Bill of Materials (BoM).
- Enable both open (market-wide) and limited (invited vendor) tender modes.
- Guarantee auditability via immutable logs, bid integrity hashing, and envelope-based price masking.
- Provide a two-stage evaluation workflow (Technical first, then Commercial) with a compliance comparison matrix.
- Deliver a full Clarification & Addenda module so vendors can ask questions and buyers can issue official amendments.
- Implement a vendor enlistment and approval workflow with document management and expiry tracking.
- Be extensible for future modules: e-Auction, reverse bidding, ERP sync, spend analytics.

2.2 Non-Goals (Phase 1)

- Automated contract signing — digital hash is used for submission integrity only.
- Integrated payment processing or escrow.
- AI-based bid evaluation — scoring is rule-based and evaluator-driven only.
- Real-time live auction / reverse bidding.
- ERP integration (SAP / Oracle PO upload).

3. User Roles & Permissions

Access control is enforced via Role-Based Access Control (RBAC). The following table summarises every role and its capabilities. A single user account may carry multiple roles (e.g., an admin who is also a buyer).

Role	Key Responsibilities	Can Access Bid Prices Before Opening?
Buyer (Purchaser)	Create & manage tenders. Define specs, features, BoM. Invite vendors. Publish addenda. Evaluate & award bids. Export results.	No — only after Bid Opening Date.
Vendor (Bidder)	Maintain company profile & capability catalog. View eligible tenders. Submit, withdraw, and resubmit bids. Respond to clarifications.	Can see only own bid values.
Evaluator	Score technical parameters. Mark compliance. Add evaluation remarks. Unlock commercial envelopes (if authorised). Cannot modify bids.	Only after commercial envelope is unlocked.
Tenant Admin	Manage users & roles within an organisation. Configure master data (Categories, UOMs, Incoterms). View organisation-level audit logs.	No.
Platform Admin	Full system access. Cross-tenant user management. System-wide audit logs. Infrastructure configuration.	Yes — for audit purposes only.

3.1 Permission Matrix — Tender Object

Action	Buyer (Owner)	Buyer (Other)	Vendor	Evaluator	Admin
Create Tender	✓	✗	✗	✗	✓
Edit Draft Tender	✓	✗	✗	✗	✓
Publish Tender	✓	✗	✗	✗	✓
View Tender Details	✓	✓	✓*	✓	✓
Submit Bid	✗	✗	✓*	✗	✗
View All Bids	✓ (post-open)	✗	✗	✓ (post-open)	✓
Score / Evaluate Bid	✗	✗	✗	✓	✗
Award Tender	✓	✗	✗	✗	✓
Cancel Tender	✓	✗	✗	✗	✓

* Vendors can view a tender only if it is Open, or if they hold an active invitation for a Limited tender.

4. Tender / RFQ Core Data Model

4.1 Tender Header — Column Definitions

Column	Type	Nullable	Default	Validation / Notes
id	UUID	No	uuid_generate_v4()	Primary key.
tender_number	TEXT	No	—	User-facing reference (e.g., RFQ-2024-001). UNIQUE per buyer_org.
buyer_org_id	UUID → orgs	No	—	FK to organisations. Must be type = 'buyer'.
title	TEXT	No	—	Max 255 characters.
tender_type	ENUM	No	—	Values: RFQ | TENDER. Drives workflow differences.
visibility	ENUM	No	—	Values: open | limited.
procurement_type	ENUM	No	—	Values: goods | works | services.
currency	CHAR(3)	No	—	ISO 4217. FK to currency_master.
price_basis	ENUM	No	unit_rate	Values: unit_rate | lump_sum. Determines how line-item totals are calculated.
fund_allocation	NUMERIC(18,2)	Yes	NULL	Buyer's budget ceiling. Hidden from vendors. Used in evaluation.
bid_security_amount	NUMERIC(18,2)	Yes	NULL	EMD amount required. NULL = no EMD required.
pre_bid_meeting_date	TIMESTAMPTZ	Yes	NULL	If set, a notification is triggered 48 h before.
pre_bid_meeting_link	TEXT	Yes	NULL	Virtual meeting URL. Visible only to invited / eligible vendors.
submission_deadline	TIMESTAMPTZ	No	—	Must be > now() at creation. Cannot be moved backward.
bid_opening_time	TIMESTAMPTZ	Yes	NULL	NULL = manual open. Otherwise auto-open at this time.
validity_days	INTEGER	No	90	Offer validity in days from submission_deadline.
status	TEXT → master	No	draft	See §4.2 State Machine.
created_by	UUID → users	No	—	The user who created the tender.
created_at	TIMESTAMPTZ	No	now()	Immutable after insert.
updated_at	TIMESTAMPTZ	No	now()	Auto-update on any row change.

4.2 Tender Status State Machine

The following transitions are the only legal moves. Any attempt to transition outside these rules returns HTTP 409 Conflict.

From Status	Allowed Next Statuses	Trigger
draft	published | cancelled	Buyer publishes or discards.
published	clarification | closed | cancelled	Vendor posts a question; deadline passes; buyer cancels.
clarification	published | closed | cancelled	Buyer answers and re-opens; deadline passes; buyer cancels.
closed	tech_eval	Bid opening occurs (auto or manual).
tech_eval	comm_eval | cancelled	All bids technically scored; buyer unlocks commercial. Or cancel.
comm_eval	awarded | cancelled	Buyer awards or cancels.
awarded	— (terminal)	No further transitions.
cancelled	— (terminal)	No further transitions.

4.3 Tender Terms (1:1 sub-table)

Column	Type	Notes
tender_id	UUID → tenders	PK & FK. Cascade delete.
payment_terms	TEXT	Free text or selected from a template library.
delivery_terms	TEXT (Incoterm)	EXW | FCA | FAS | FOB | CFR | CIF | CIP | DAP | DAT | DDP.

5. Line Items & Bill of Materials (BoM)

5.1 tender_items — Hierarchical Structure

Items form a tree via parent_item_id. A row where item_type = 'group' is a logical container (its quantity is informational only). Leaf rows (item_type = 'item') carry the quantity that vendors must price. The sl_no uniqueness is scoped to (tender_id, parent_item_id) to allow independent numbering per group.

Column	Type	Nullable	Notes
id	UUID	No	PK.
tender_id	UUID → tenders	No	FK. Cascade delete.
parent_item_id	UUID → self	Yes	NULL = root item. Non-null = child of another item.
item_type	ENUM	No	group | item. Groups cannot be priced by vendors.
sl_no	INTEGER	No	Display sequence. UNIQUE per (tender_id, parent_item_id).
item_code	TEXT	Yes	Buyer's internal SKU. Not visible to vendors if configured.
item_name	TEXT	No	Short label. Max 200 chars.
specification	TEXT	Yes	Rich-text spec. Supports markdown or stored HTML.
quantity	NUMERIC(14,3)	No	0 for group rows. Must be > 0 for item rows.
uom	TEXT → uom_master	No	FK to unit-of-measure master.
estimated_cost	NUMERIC(18,2)	Yes	Buyer's hidden budget per unit. Never exposed to vendors.
created_at	TIMESTAMPTZ	No	Auto.

5.2 Feature & Selection Engine

Buyers attach structured feature groups to each line item. Vendors must respond to every mandatory feature. This replaces unstructured free-text bids with machine-comparable data.

feature_definitions — Master catalogue of reusable features

Column	Type	Notes
id	UUID	PK.
name	TEXT	e.g., 'Processor', 'Warranty Period'. Max 100 chars.
feature_type	ENUM	single_select | multi_select | text | numeric | boolean.
scoring_type	ENUM	binary (compliant/not) | graded (score per option) | numeric (evaluator assigns score).
evaluation_weight	NUMERIC(5,2)	Weight in technical scoring (0.00–100.00). Sum across features per item should = 100.
is_global	BOOLEAN	If TRUE, this feature can be reused across tenders without re-creation.

feature_options — Selectable values for a feature

Column	Type	Notes
id	UUID	PK.
feature_id	UUID → feature_definitions	FK. Cascade delete.
option_value	TEXT	Display label, e.g., 'Intel i7'. Max 150 chars.
option_score	NUMERIC(5,2)	Pre-assigned score for graded scoring_type. 0 for binary.
sort_order	INTEGER	Display order within the feature.

tender_item_features — Attaches features to a specific line item

Column	Type	Notes
tender_item_id	UUID → tender_items	Composite PK part 1.
feature_id	UUID → feature_definitions	Composite PK part 2.
is_mandatory	BOOLEAN	TRUE = vendor must respond. FALSE = optional bonus.

6. Vendor Bid Submission & Versioning

Bids are encrypted at rest (AES-256). No one — including Platform Admin — can read bid content until the bid_opening_time has passed or the buyer manually opens bids. Versioning allows vendors to withdraw and resubmit without losing history.

6.1 bids — Bid Header

Column	Type	Nullable	Notes
id	UUID	No	PK.
tender_id	UUID → tenders	No	FK.
vendor_org_id	UUID → orgs	No	FK. The vendor organisation.
version	INTEGER	No	Starts at 1. Incremented on each resubmission. UNIQUE per (tender_id, vendor_org_id, version).
status	ENUM	No	draft | submitted | withdrawn. See §6.3.
total_amount	NUMERIC(18,2)	Yes	Auto-calculated sum of all bid_items totals + taxes. Stored for fast comparison.
compliance_status	ENUM	Yes	compliant | non_compliant | partial. Derived after evaluation but stored for quick filtering.
digital_hash	TEXT	Yes	SHA-256 of the bid payload at submission time. Used for integrity verification.
submitted_at	TIMESTAMPTZ	Yes	Exact timestamp of submission. NULL while in draft.
created_at	TIMESTAMPTZ	No	Row creation time.
updated_at	TIMESTAMPTZ	No	Auto-update.

6.2 bid_envelopes — Technical / Commercial Separation

Each bid is split into envelopes. The technical envelope is opened first; the commercial envelope remains sealed until the evaluator (or system) explicitly unlocks it for technically qualified bids only.

Column	Type	Notes
id	UUID	PK.
bid_id	UUID → bids	FK.
envelope_type	ENUM	technical | commercial. One row per type per bid.
is_open	BOOLEAN	FALSE until explicitly opened.
opened_at	TIMESTAMPTZ	Nullable. Set when is_open flips to TRUE.
opened_by	UUID → users	Nullable. The user who opened it (for manual open).

6.3 Bid Versioning State Machine

From Status	Allowed Next	Trigger / Rule
draft	submitted | withdrawn	Vendor finalises and submits before deadline, or discards draft.

submitted	withdrawn	Vendor withdraws before submission_deadline. Creates no new row.
withdrawn	— (terminal for this version)	A new version row is created when the vendor submits again.

Active bid resolution: SELECT * FROM bids WHERE tender_id = X AND vendor_org_id = Y AND status = 'submitted' ORDER BY version DESC LIMIT 1. If no row exists, the vendor has no active bid.

6.4 bid_items — Per-Line-Item Vendor Response

Column	Type	Nullable	Notes
id	UUID	No	PK.
bid_id	UUID → bids	No	FK.
tender_item_id	UUID → tender_items	No	FK. Must exist in the referenced tender.
envelope_type	ENUM	No	technical | commercial. Determines which envelope this data belongs to.
unit_price	NUMERIC(18,4)	Yes	Vendor's unit price. NULL in the technical envelope.
item_total_price	NUMERIC(18,2)	Yes	Cached: quantity × unit_price. Recalculated on read if NULL.
compliance	ENUM	Yes	compliant | non_compliant | partial.
non_compliance_remarks	TEXT	Yes	Mandatory if compliance ≠ compliant.
brand_make	TEXT	Yes	Vendor's brand / make for this item.

6.5 bid_item_feature_values — Vendor Feature Selections

Column	Type	Notes
bid_item_id	UUID → bid_items	Composite PK part 1.
feature_id	UUID → feature_definitions	Composite PK part 2.
option_id	UUID → feature_options	Set for single_select / multi_select. NULL for text / numeric.
text_value	TEXT	Set for feature_type = text. NULL otherwise.
numeric_value	NUMERIC(12,3)	Set for feature_type = numeric. NULL otherwise.

Validation rule: For every mandatory feature attached to a tender_item, the vendor MUST have a corresponding row in bid_item_feature_values. Submission is rejected otherwise.

7. Tender Visibility & Invitation Lifecycle

7.1 Open Tender

- Visible to all registered and approved vendors.
- Any vendor passing the tender's qualification requirements (§8) may submit a bid.
- No invitation token is needed.

7.2 Limited Tender — Invitation System

The buyer selects vendors from the enlisted & approved vendor list. The system generates a unique, cryptographically random invitation token per vendor. The token is embedded in the notification link.

tender_vendor_invitations

Column	Type	Nullable	Notes
tender_id	UUID → tenders	No	Composite PK part 1.
vendor_org_id	UUID → orgs	No	Composite PK part 2.
invitation_token	TEXT	No	Cryptographically random (min 32 bytes, hex-encoded). Single-use for first access verification.
status	ENUM	No	sent | viewed | bid_started | submitted | declined. See state machine below.
invited_at	TIMESTAMPTZ	No	When the invitation was created.
viewed_at	TIMESTAMPTZ	Yes	First time the vendor opened the tender via the token link.
declined_at	TIMESTAMPTZ	Yes	If vendor explicitly declines.
declined_reason	TEXT	Yes	Optional free-text reason for declining.

Invitation Status State Machine

From	To	Trigger
sent	viewed | declined	Vendor clicks link / vendor explicitly declines.
viewed	bid_started | declined	Vendor begins drafting a bid / declines.
bid_started	submitted | declined	Vendor submits bid / declines before deadline.
submitted	— (terminal)	Bid is in the system.
declined	— (terminal)	No further action from this vendor.

8. Vendor Enlistment & Approval Workflow

Vendors must complete enlistment before they can participate in any tender. The buyer organisation controls approval. Document expiry is tracked and alerts are sent automatically.

8.1 vendor_profiles

Column	Type	Nullable	Notes
organization_id	UUID → orgs	No	PK. 1:1 with organisations where type = 'vendor'.
legal_name	TEXT	No	Full legal entity name.
tax_id	TEXT	Yes	VAT / GST / TIN number.
contact_name	TEXT	Yes	Primary point of contact.
contact_email	TEXT	Yes	Primary contact email.
contact_phone	TEXT	Yes	Primary phone.
website	TEXT	Yes	Company website URL.
status	ENUM	No	pending | approved | rejected | suspended.
status_changed_at	TIMESTAMPTZ	No	When status last changed.
status_changed_by	UUID → users	Yes	The buyer/admin who changed status.
rejection_reason	TEXT	Yes	Mandatory if status = rejected.
created_at	TIMESTAMPTZ	No	Auto.

8.2 vendor_categories — Capability Mapping

Column	Type	Notes
vendor_org_id	UUID → orgs	Composite PK part 1.
category_id	UUID → categories_master	Composite PK part 2. Links to tender categories.
added_at	TIMESTAMPTZ	Auto.

8.3 vendor_documents — Compliance Documents

Column	Type	Nullable	Notes
id	UUID	No	PK.
vendor_org_id	UUID → orgs	No	FK.
document_type	ENUM	No	trade_license | vat_certificate | iso_cert | other.
file_url	TEXT	No	Object storage path (S3 / MinIO).
issued_date	DATE	Yes	Date the document was issued.
expiry_date	DATE	Yes	Date the document expires. Triggers alerts 30 / 7 days before.

uploaded_at	TIMESTAMPTZ	No	Auto.
uploaded_by	UUID → users	No	The user who uploaded.

8.4 Enlistment Status State Machine

From	Allowed Next	Trigger
pending	approved | rejected	Buyer reviews and approves or rejects.
approved	suspended	Buyer suspends (temporary ban).
rejected	pending	Vendor re-applies after addressing concerns.
suspended	approved | rejected	Buyer lifts suspension or formally rejects.

9. Clarifications & Addenda

9.1 clarification_questions

Vendors can post questions against a published tender. Visibility is configurable: either only the buyer sees the question, or all invited/eligible vendors see it (to promote transparency).

Column	Type	Nullable	Notes
id	UUID	No	PK.
tender_id	UUID → tenders	No	FK.
vendor_org_id	UUID → orgs	No	The vendor who asked.
question_text	TEXT	No	The question. Max 2000 chars.
is_public	BOOLEAN	No	TRUE = all vendors see Q&A. FALSE = only buyer sees.
status	ENUM	No	open | answered | closed.
created_at	TIMESTAMPTZ	No	Auto.

9.2 clarification_answers

Column	Type	Notes
id	UUID	PK.
question_id	UUID → questions	FK. One answer per question.
buyer_user_id	UUID → users	The buyer user who answered.
answer_text	TEXT	Official answer. Max 5000 chars.
creates_addendum	BOOLEAN	If TRUE, the buyer must also publish a formal Addendum.
answered_at	TIMESTAMPTZ	Auto.

9.3 addenda — Official Amendments

Column	Type	Nullable	Notes
id	UUID	No	PK.
tender_id	UUID → tenders	No	FK.
addendum_number	INTEGER	No	Sequential within a tender (1, 2, 3…).
title	TEXT	No	Short title of the amendment.
description	TEXT	No	Full description of changes.
extends_deadline_days	INTEGER	Yes	If > 0, the system auto-extends submission_deadline by this many days.
published_at	TIMESTAMPTZ	No	When the addendum went live.
published_by	UUID → users	No	Buyer user.

9.4 addendum_acknowledgements — Vendor Acknowledgement

Column	Type	Nullable	Notes
addendum_id	UUID → addenda	No	Composite PK part 1.
vendor_org_id	UUID → orgs	No	Composite PK part 2.
acknowledged_at	TIMESTAMPTZ	Yes	NULL = not yet acknowledged. Set on vendor action.
acknowledged_by	UUID → users	Yes	The vendor-side user who clicked 'Acknowledge'.

Business rule: A vendor cannot submit a bid if any addendum on that tender remains unacknowledged by their organisation.

10. Evaluation & Comparison Module

10.1 Two-Stage Evaluation Workflow

- Stage 1 — Technical: Commercial envelopes remain sealed. Evaluators score each bid's features and compliance per line item. Output: a list of technically qualified bids (those meeting a configurable minimum technical score).
- Stage 2 — Commercial: The buyer or authorised evaluator unlocks the commercial envelope for every technically qualified bid. Prices become visible. The system ranks bids by total price (L1 basis) or evaluator-weighted scoring.

10.2 evaluations — Scoring Table

Column	Type	Nullable	Notes
id	UUID	No	PK.
bid_id	UUID → bids	No	FK.
evaluator_id	UUID → users	No	FK. The evaluator.
technical_score	NUMERIC(6,2)	Yes	Weighted sum of feature scores. Range 0–100.
commercial_score	NUMERIC(6,2)	Yes	Price-based score. Calculated by system after commercial unlock.
overall_score	NUMERIC(6,2)	Yes	Final blended score: (tech_weight × technical_score) + (comm_weight × commercial_score).
is_technically_qualified	BOOLEAN	Yes	TRUE if technical_score ≥ tender's min_technical_score threshold.
remarks	TEXT	Yes	Free-text evaluation notes.
created_at	TIMESTAMPTZ	No	Auto.
updated_at	TIMESTAMPTZ	No	Auto.

10.3 evaluation_line_scores — Per-Item Granular Scoring

Column	Type	Notes
evaluation_id	UUID → evaluations	Composite PK part 1.
tender_item_id	UUID → tender_items	Composite PK part 2.
feature_id	UUID → feature_defs	Composite PK part 3.
score	NUMERIC(5,2)	Score assigned by evaluator for this feature on this item.
remarks	TEXT	Optional per-feature remark.

10.4 Comparison Matrix Logic

- Compliance Matrix: For each (vendor × line_item), display compliant (green) | partial (yellow) | non_compliant (red).
- Feature Comparison: Side-by-side table showing each vendor's selected option per feature.

- Financial Comparison: Row = line item, Column = vendor. Shows unit_price and item_total_price. Footer row = grand total including taxes.
- Ranking: Bids are ranked by overall_score (descending). Ties broken by total_amount (ascending).

11. Award Management

Awards can be full (entire tender to one vendor) or partial (different line items to different vendors). This supports split-award procurement common in government and large-scale EPC projects.

11.1 awards

Column	Type	Nullable	Notes
id	UUID	No	PK.
tender_id	UUID → tenders	No	FK. Denormalized for fast query.
tender_item_id	UUID → tender_items	No	FK. The line item being awarded.
bid_id	UUID → bids	No	FK. The winning bid for this item.
awarded_quantity	NUMERIC(14,3)	No	Quantity awarded. Must be ≤ tender_item quantity.
awarded_price	NUMERIC(18,2)	No	Agreed unit price for the award.
awarded_at	TIMESTAMPTZ	No	Auto.
awarded_by	UUID → users	No	Buyer user who issued the award.

- Business rule: Once a tender is awarded, its status moves to 'awarded'. No further bid submissions or evaluations are permitted.
- Business rule: The sum of awarded_quantity across all awards for a tender_item must not exceed that item's original quantity.

12. Notifications & Deadline Management

12.1 Notification Channels

- Email (primary — always sent).
- SMS (optional — enabled per organisation or per tender).
- In-App Toast (real-time push via WebSocket or SSE).

12.2 notifications — Tracking Table

Column	Type	Nullable	Notes
id	UUID	No	PK.
tender_id	UUID → tenders	Yes	Nullable — some notifications are user-level only.
recipient_id	UUID → users	No	The user who receives this notification.
notification_type	ENUM	No	See §12.3 for full list.
channel	ENUM	No	email | sms | in_app.
status	ENUM	No	pending | sent | delivered | failed | retried.
payload	JSONB	Yes	Notification-specific data (e.g., tender title, deadline).
sent_at	TIMESTAMPTZ	Yes	When the message was dispatched.
failed_reason	TEXT	Yes	Error detail if status = failed.
created_at	TIMESTAMPTZ	No	Auto.

12.3 Notification Type Catalogue

Type	Trigger	Recipients
tender_published	Tender status → published.	All eligible / invited vendors.
invitation_sent	Invitation row created.	The invited vendor.
addendum_published	New addendum row inserted.	All invited / eligible vendors.
addendum_unacknowledged	24 h before deadline; addendum still not acknowledged.	Vendor contacts with pending acks.
clarification_answered	Buyer posts an answer.	The asking vendor (+ all if public).
deadline_reminder_3d	3 days before submission_deadline.	All vendors who haven't submitted.
deadline_reminder_1d	1 day before submission_deadline.	All vendors who haven't submitted.
bid_submitted	Vendor's bid status → submitted.	The vendor + the buyer.
bid_opening	Bids opened (auto or manual).	Buyer + all evaluators.
tender_awarded	Award row inserted.	All participating vendors + buyer.
vendor_doc_expiry_3	30 days before a	Vendor contacts + tenant admin.

0d	vendor_document expiry_date.	
vendor_doc_expiry_7d	7 days before expiry.	Vendor contacts + tenant admin.

12.4 Retry Policy

- Email / SMS: Up to 3 retries with exponential back-off (1 min, 5 min, 25 min). After 3 failures, status = failed; an alert is logged.
- In-App: Delivered via real-time channel only. No retry — the notification remains in the user's notification centre until dismissed.

13. Audit, Security & Data Integrity

13.1 Immutable Audit Log

The audit_logs table is append-only. The application layer must never execute UPDATE or DELETE against it. At the database level, revoke these privileges from the application service account.

Column	Type	Notes
id	UUID	PK.
actor_id	UUID → users	The user who performed the action. NULL for system-generated events.
action	TEXT	Categorised action code, e.g., TENDER_CREATED, BID_SUBMITTED, ENVELOPE_OPENED.
entity_type	TEXT	The affected entity: tender | bid | vendor | user | addendum.
entity_id	UUID	The PK of the affected entity.
metadata	JSONB	Additional context. E.g., {old_status, new_status} for status changes.
created_at	TIMESTAMPTZ	Auto. Immutable.

13.2 Encryption

- At Rest: AES-256 encryption applied to all columns in bid_items and bid_item_feature_values that contain pricing or vendor-identifying data. Encryption keys are managed per-tenant and rotated quarterly.
- In Transit: TLS 1.3 enforced on all API endpoints. HSTS header with min-age = 31536000.

13.3 Bid Integrity Verification

- On submission, the backend computes SHA-256(JSON payload of the bid) and stores it in bids.digital_hash.
- During evaluation, the system recomputes the hash and compares it to the stored value. A mismatch flags the bid as TAMPERED and blocks evaluation.
- A "Bid Integrity Report" can be exported by the buyer showing hash, computation timestamp, and verification status for every bid.

13.4 Rate Limiting & DDoS Protection

- Bid submission endpoint: Max 5 requests per 10 seconds per vendor_org_id. Excess → HTTP 429.
- Login endpoint: Max 10 attempts per 15 minutes per IP. Excess → 15-minute lockout.
- General API: 200 requests per minute per authenticated user. Configurable per tenant.

14. Qualification & Eligibility Requirements

Buyers can attach pre-qualification criteria to a tender. Vendors must respond to each requirement during bid submission. Mandatory requirements that are not met disqualify the bid automatically.

14.1 tender_qualification_requirements

Column	Type	Notes
id	UUID	PK.
tender_id	UUID → tenders	FK.
requirement	TEXT	Description of the requirement, e.g., 'Minimum 5 years of experience in similar projects'.
is_mandatory	BOOLEAN	TRUE = must be compliant to qualify. FALSE = scored but not disqualifying.

14.2 bid_qualification_responses

Column	Type	Notes
requirement_id	UUID → requirements	Composite PK part 1.
bid_id	UUID → bids	Composite PK part 2.
response	TEXT	Vendor's response / evidence description.
document_id	UUID → attachments	Nullable. Supporting document uploaded by vendor.
compliant	BOOLEAN	Evaluator marks TRUE/FALSE after review.

15. Tax & Multi-Currency Engine

Tenders are denominated in a single currency. Tax is calculated per line item based on configurable tax rules. Multi-currency conversion rates are fetched from an external API daily and cached.

15.1 tax_rules — Configurable Tax Catalogue

Column	Type	Notes
id	UUID	PK.
name	TEXT	e.g., 'GST 18%', 'VAT 10%'.
rate_percent	NUMERIC(6,2)	Tax rate, e.g., 18.00.
applies_to	ENUM	goods | works | services | all. Must match tender procurement_type or 'all'.
is_active	BOOLEAN	FALSE = no longer applied to new tenders.
created_at	TIMESTAMPTZ	Auto.

15.2 bid_item_taxes — Per-Item Tax Breakdown

Column	Type	Notes
bid_item_id	UUID → bid_items	Composite PK part 1.
tax_rule_id	UUID → tax_rules	Composite PK part 2.
tax_amount	NUMERIC(18,2)	Calculated: item_total_price × (rate_percent / 100).

15.3 currency_rates — Daily FX Cache

Column	Type	Notes
base_currency	CHAR(3)	Always the platform's base currency (e.g., USD).
target_currency	CHAR(3)	The target ISO code.
rate	NUMERIC(12,6)	1 unit of base = rate units of target.
fetched_at	TIMESTAMPTZ	When this rate was retrieved.

16. Export & Reporting Engine

Buyers and evaluators can export tender data in structured formats for offline review, archival, or integration with ERP systems.

16.1 Supported Exports

Export Type	Format	Scope	Who Can Export
Tender Summary	PDF	Single tender header + terms.	Buyer (owner), Admin.
Bid Comparison Matrix	PDF / XLSX	All bids for a tender — compliance + pricing.	Buyer, Evaluator, Admin.
Bid Integrity Report	PDF	Hash verification for all bids.	Buyer, Admin.
Award Letter	PDF	Per-vendor award detail.	Buyer (owner), Admin.
Full Tender Data Dump	XLSX	All items, features, bids, scores — raw data.	Admin only.

16.2 Implementation Note

- PDF generation is handled asynchronously via the task queue (BullMQ / RabbitMQ). The user receives a download link via notification once the export is ready.
- XLSX exports use a streaming row-writer to handle tenders with thousands of line items without memory issues.

17. Non-Functional Requirements

17.1 Performance

Metric	Target	Measurement
Page Load (P95)	< 2 seconds	From navigation start to interactive.
Bid Submission ACK	< 1 second	From client POST to 200 response.
Concurrent Users	10,000+	During peak closing-minute window.
DB Query (single tender)	< 200 ms	Full tender + items + features read.
Search Latency	< 300 ms	Tender / vendor full-text search.

17.2 Availability & Reliability

- SLA: 99.9% uptime (< 8.77 hours downtime per year).
- Automated daily backups with point-in-time recovery (PITR) up to 7 days.
- Health-check endpoint: GET /health returns 200 if DB + cache + queue are reachable.

17.3 Scalability & Multi-Tenancy

- Row-Level Security (RLS) in PostgreSQL enforces tenant isolation. Each tenant's data is invisible to others at the query level.
- Stateless API layer — sessions stored in Redis. Horizontal scaling via container orchestration (Kubernetes / ECS).
- JSONB indexes on feature columns for sub-millisecond feature-based filtering.

17.4 Security

- Passwords: bcrypt with cost factor ≥ 12.
- API Auth: Short-lived JWT (15 min) + refresh token rotation.
- CORS: Whitelist only the platform's frontend origin.
- CSP headers on all responses.
- SQL injection prevention: parameterised queries only — no string interpolation.

18. Suggested Technology Stack

Layer	Recommended	Rationale
Frontend	Next.js (React) + Ant Design	SSR for SEO on public tender pages. Ant Design's Table & Form components match the data-heavy UI.
Backend	Node.js (NestJS)	Modular structure maps cleanly to the domain modules. TypeScript end-to-end.
Database	PostgreSQL 16+	JSONB for features, RLS for multi-tenancy, PITR for recovery.
Cache / Session	Redis 7+	Session store, rate-limit counters, queue locking.
Search	Meilisearch	Near-instant search on tenders and vendors. Simpler ops than Elasticsearch for this scale.
Object Storage	AWS S3 / MinIO	Attachments, exports, vendor documents. Presigned URLs for direct upload.
Task Queue	BullMQ (Redis-backed)	Async jobs: notifications, PDF export, hash computation.
Email	AWS SES / SendGrid	Reliable delivery at scale. Webhook for delivery receipts.
Monitoring	Prometheus + Grafana	Dashboards for latency, error rates, queue depth.

19. Future Enhancements (Post Phase 1)

Feature	Description	Schema Impact
Reverse Auction	Live bidding where vendors compete in real-time to lower price.	New: auction_sessions, auction_bids tables. WebSocket support.
e-Contract Generation	Auto-generate contract PDF from award data. DocuSign integration for signing.	New: contracts table. Integration config table.
ERP Integration	Sync awarded POs to SAP / Oracle via API.	New: erp_sync_jobs table. Webhook config.
Spend Analytics	Dashboard showing spend by category, vendor, time period.	Read-only views / materialised views on awards + bids.
AI Anomaly Detection	Flag bids with pricing outliers or collusion patterns.	New: anomaly_flags table. ML model output stored as JSONB.
Vendor Portal Mobile	Dedicated mobile app for vendors to track invitations and submit bids.	No schema change — API layer only.